Brief Report

Folia Primatol 2001;72:26–29

Received: July 24, 2000
Accepted after revision: October 15, 2000

Spectral Sensitivity of Gibbons: Implications for Photopigments and Color Vision

Jess F. Deegan II[a] Gerald H. Jacobs[b]

[a]California State University, Bakersfield, Calif., and [b]University of California, Santa Barbara, Calif., USA

Key Words
Gibbon · Photopigments · Spectral sensitivity · Color vision

Introduction

In an article published in the inaugural volume of this journal, Tigges [1] reported results from tests of an orang-utan *(Pongo pygmaeus)* and three gibbons *(Hylobates lar)* that were trained to discriminate colored papers from among a series of gray papers that varied widely in reflectance. Although the orang-utan learned to discriminate successfully each of four colored papers (having human hue designations of blue, green, yellow and red) from the grays, the results from the gibbons were more variable. In particular, all three subjects showed some difficulty in discriminating between blue and some of the gray papers. This result might suggest that the color vision of gibbons is different from that of orang-utans and humans.

In the past 20 years the genes specifying the cone photopigment proteins have been extensively investigated and, as well, the interrelationships between these genes, the photopigments they produce, and the manner in which these photopigments are linked to color vision capacities have been worked out in some detail. Relevant here is the fact that among mammals only primates have trichromatic color vision [2]. In catarrhine primates, trichromacy emerged some 35 MYA following an X-chromosome opsin gene duplication that left separate genes specifying the middle- (M) and long-wavelength (L)-sensitive cone pigments [3]. In conjunction with an autosomal gene that yields a short (S)-wavelength-sensitive pigment, this provides three classes of cone photoreceptors and consequent trichromatic color vision. It appears that all subsequent catarrhine primates have conservatively maintained this basic trichromatic arrangement. Thus,

KARGER

Fax + 41 61 306 12 34
E-Mail karger@karger.ch
www.karger.com

© 2001 S. Karger AG, Basel
0015–5713/01/0721–0026$17.50/0

Accessible online at:
www.karger.com/journals/fpr

Gerald H. Jacobs
Neuroscience Research Institute
University of California
Santa Barbara, CA 93106 (USA)
Fax +1 805 893 2005, E-Mail jacobs@psych.ucsb.edu

behavioral tests of color vision, measurements of photopigments and examination of cone opsin genes in a wide variety of Old World monkeys and great apes reveal a picture consistent with the presence of trichromatic color vision capacities that show little variation across species [4–12]. All of this suggests that gibbons too should enjoy full trichromatic color vision. Here we report measurements of gibbon spectral sensitivity that strongly support that expectation.

Materials and Methods

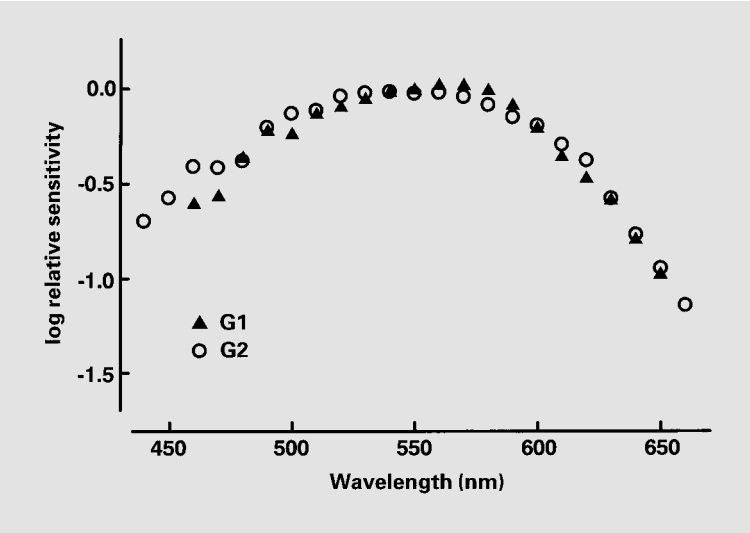
A gross potential, the electroretinogram (ERG), was used to measure spectral sensitivity. The procedure, and its application to primate subjects, has been fully described [13]. In brief, signals were recorded from a contact lens electrode placed on the eye of a sedated animal. The pupil was dilated with a mydriatic and the cornea was topically anesthetized. ERGs were elicited by monochromatic lights flickered (31.25 Hz) onto the retina as 59-degree circular fields. Sensitivity was measured by equating the responses to such lights and to interspersed, similarly flickering, achromatic lights using a flicker photometric procedure. Sensitivity measurements were made at 10-nm steps from 450 to 660 nm. We also used chromatic adaptation to test for response univariance in the middle to long wavelengths. For this, a photometric equation was made for two test stimuli (540 and 630 nm). These equations were made in a neutrally adapted eye and then, subsequently, as the eye was continuously adapted to 540- and 630-nm lights. The logic is that, if there is only a single photopigment active over this spectral range, the equation will be independent of the adaptation state, but the presence of two photopigments means the equations will change systematically depending on whether the eye is adapted to red or to green light [14].

A male (10 years old) and a female (14 years old) gibbon *(H. lar)* were studied during the course of routine physical examinations. Animals were sedated by intramuscular injection of ketamine hydrochloride, intubated and maintained on 3% isoflurane during the recordings.

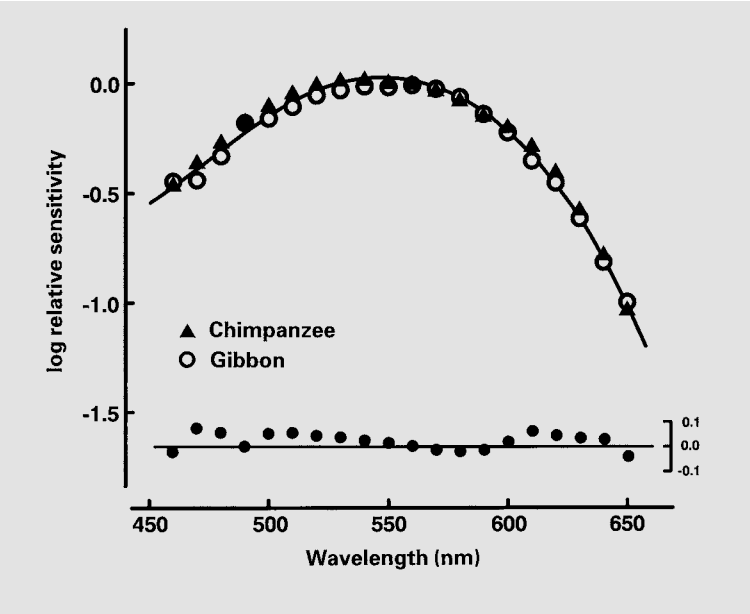
Results and Discussion

We found clear evidence for differential chromatic adaptation of M and L cones in both subjects. In earlier studies it proved convenient to express the magnitude of this effect by calculating the differences (in log units) in the photometric equation values determined with 540- and 630-nm adaptation. So specified, a subject with only a single M/L pigment gives a value of 0 as, for instance, human dichromats. The two gibbons had values of 0.073 and 0.14. These are within the range of values determined using the same test for 26 common chimpanzees [12], a species known to have trichromatic color vision.

The spectral sensitivity functions obtained for the two gibbons are shown in figure 1. With the exception of one aberrant point (460 nm), the two have very similar sensitivity at all wavelengths. Peak sensitivity is at about 560 nm and for the 16 test wavelengths that were run in common the average difference in sensitivity between the two animals amounted to only 0.073 log unit (SD = 0.07). The M and L photopigment complements of the two gibbons must be identical or very nearly so. Figure 2 compares the spectral sensitivity of gibbons and chimpanzees. The inset at the bottom of figure 2 shows the absolute differences in sensitivity between the two species. These differences are very small and nonsystematic across wavelengths, suggesting the M and L cone pigments of the two apes are likely to be the same. On that assumption, the spectral sensitivity curves have been best fit by linearly summing standard photopigment absorption curves having peak values known to be accurate for the chimpanzee [12].



1



2

Fig. 1. Spectral sensitivity functions for two gibbons obtained from flicker photometric ERG measurements. The functions for the two (G1 = male; G2 = female) have been shifted on the sensitivity axis so as to have the same average sensitivity.

Fig. 2. Mean spectral sensitivity functions obtained from 26 common chimpanzees and 2 gibbons. The functions for the two were shifted on the sensitivity axis to have the same average sensitivities. The data for each have been corrected by assuming they share a common lens absorbance. The wavelength-by-wavelength differences in sensitivity (specified in log units) for the two species are plotted in the inset at the bottom. The continuous curve fitted to the results from chimpanzees and gibbons is the best fitting linear summation of M (peak = 530 nm) and L (peak = 563 nm) photopigment absorption curves in the proportions 530 (44%) + 563 (56%).

In sum, spectral sensitivity functions measured using an electrophysiological procedure were nearly identical for two gibbons. These results provide evidence for the presence of separate M and L photopigments that seem virtually indiscriminable from those of a known trichromat, the common chimpanzee. We conclude that gibbons maintain the same basic photopigment complement as other catarrhine primates and should, accordingly, have trichromatic color vision very similar to that of other representatives of this group.

Acknowledgment

We thank C. Sweet and L. Boldrick of the Santa Ana Zoo for their cooperation and help. This work was supported by grant EY02052 from the National Eye Institute.

References

1 Tigges J: On color vision in gibbon and orang-utan. Folia Primatol 1963;1:188–198.
2 Jacobs GH: The distribution and nature of colour vision among the mammals. Biol Rev 1993;68:413–471.
3 Nathans J, Thomas D, Hogness DS: Molecular genetics of human color vision: The genes encoding blue, green and red pigments. Science 1986;232:193–202.
4 De Valois RL, Morgan HC, Polson MC, Mead WR, Hull EM: Psychophysical studies of monkey vision. I. Macaque luminosity and color vision tests. Vision Res 1974;14:53–67.
5 Jacobs GH, Neitz J, Crognale MA, Brammer GT: Spectral sensitivity of vervet monkeys *(Cercopithecus aethiops sabaeus)* and the issue of catarrhine trichromacy. Am J Primatol 1991;23:185–195.
6 Bowmaker JK, Astell S, Hurst DM, Mollon JD: Photosensitive and photostable pigments in the retinae of Old World monkeys. J Exp Biol 1991;156:1–19.
7 Hunt DM, Cowing JA, Patel R, Appukuttan B, Bowmaker JK, Mollon JD: Sequence and evolution of the blue cone pigment gene in Old and New World primates. Genomics 1995;27:535–538.
8 Jacobs GH, Deegan JF II: Spectral sensitivity of macaque monkeys measured with ERG flicker photometry. Vis Neurosci 1997;14:921–928.
9 Jacobs GH, Deegan JF II: Uniformity of colour vision in Old World monkeys. Proc R Soc Lond B 1999;266: 2023–2028.
10 Deeb SS, Jorgensen AL, Battisti L, Iwasaki L, Motulsky AG: Sequence divergence of the red and green visual pigments in great apes and humans. Proc Natl Acad Sci USA 1994;91:7262–7266.
11 Dulai KS, Bowmaker JK, Mollon JD, Hunt DM: Sequence divergence, polymorphism and evolution of middle-wave and long-wave visual pigment genes of great apes and Old World monkeys. Vis Res 1994;34:2483–2491.
12 Jacobs GH, Deegan JF II, Moran JL: ERG measurements of spectral sensitivity of common chimpanzees *(Pan troglodytes)*. Vision Res 1996;36:2587–2594.
13 Jacobs GH, Neitz J, Krogh K: Electroretinogram flicker photometry and its applications. J Opt Soc Am A 1996; 13:641–648.
14 Jacobs GH, Neitz J: Inheritance of color vision in a New World monkey *(Saimiri sciureus)*. Proc Natl Acad Sci USA 1987;84:2545–2549.